



University
of Rochester



Principled preference-guided multi-objective learning

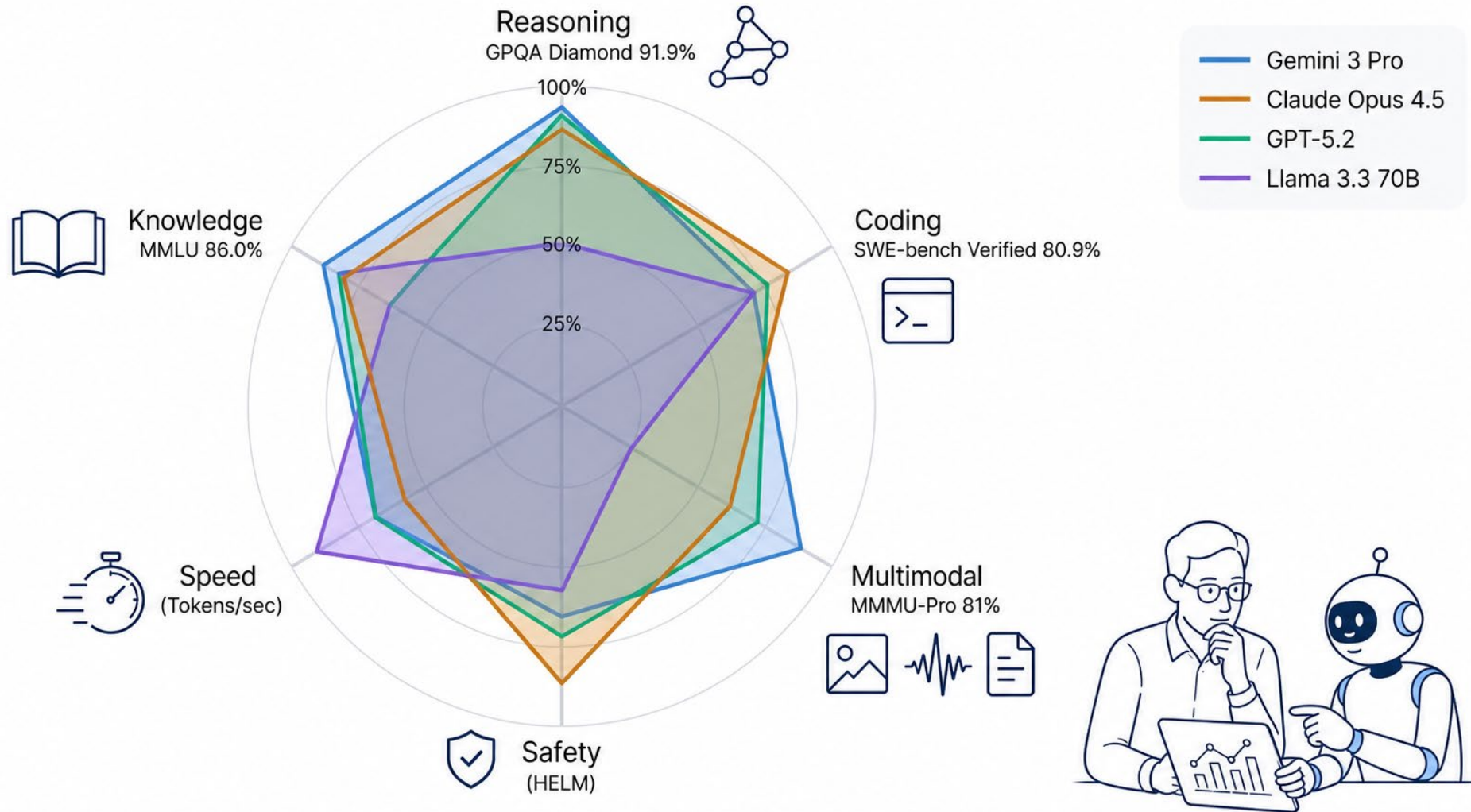
--- uniformly traversing the Pareto front

Presenter: Lisha Chen
University of Rochester

Multi-objective learning (MOL) is common



AI capabilities measured by multi-tasking



Sources: GPQA Diamond, SWE-bench Verified, MMMU-Pro, HELM Safety, MMLU (Dec 2025)

LLM score vs cost Pareto Frontier curve

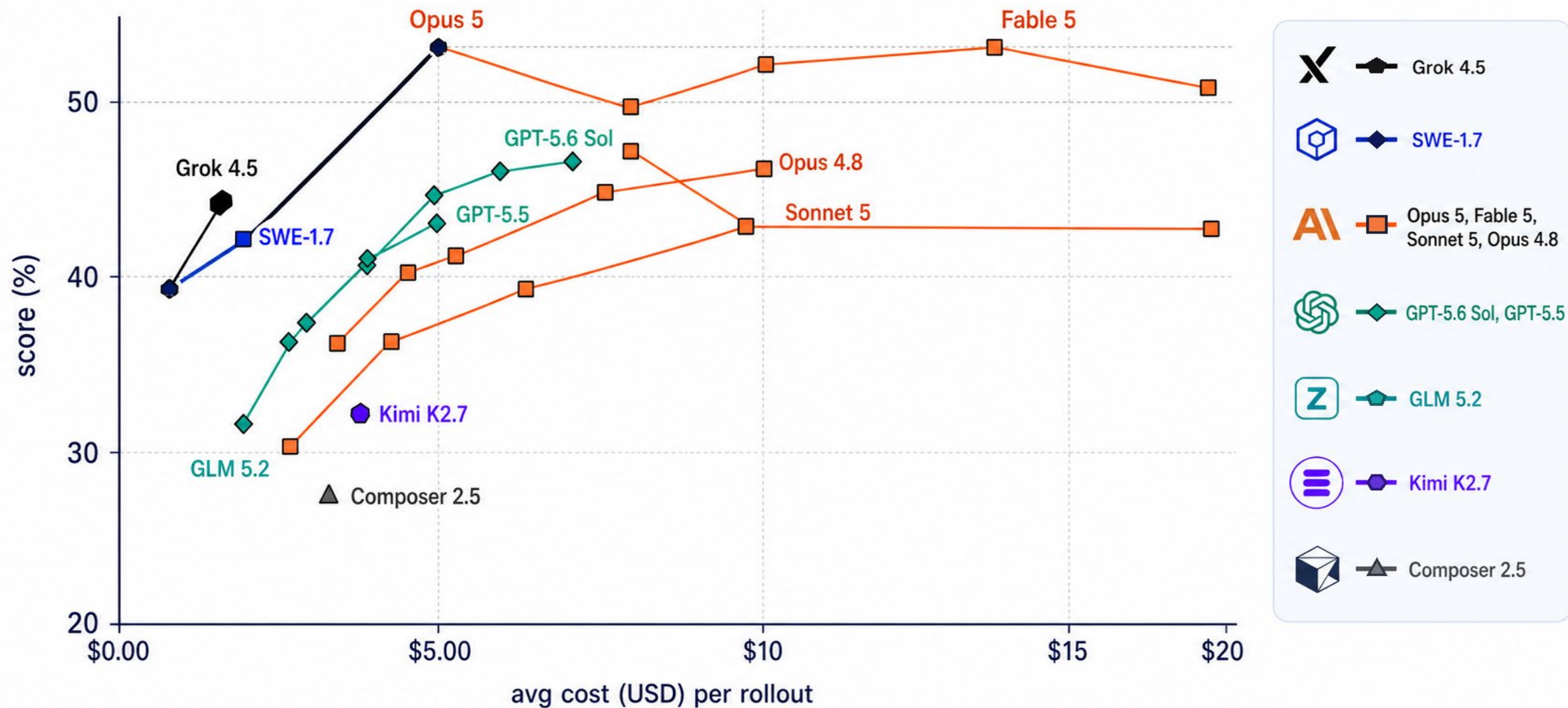


Figure from **FrontierCode 1.1 Leaderboard**

Pareto optimality, Pareto front

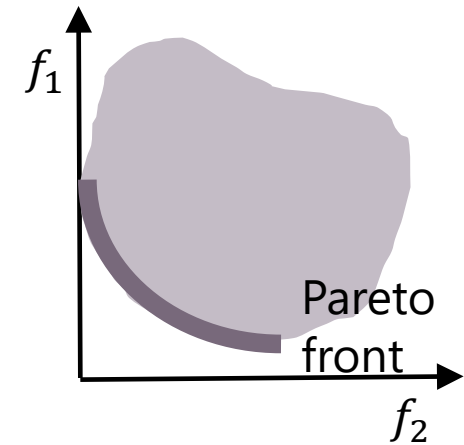
$$\text{MOL} \quad \min_{x \in X} F(x) = [f_1(x); \dots; f_M(x)]$$

Definition (weak Pareto optimal)

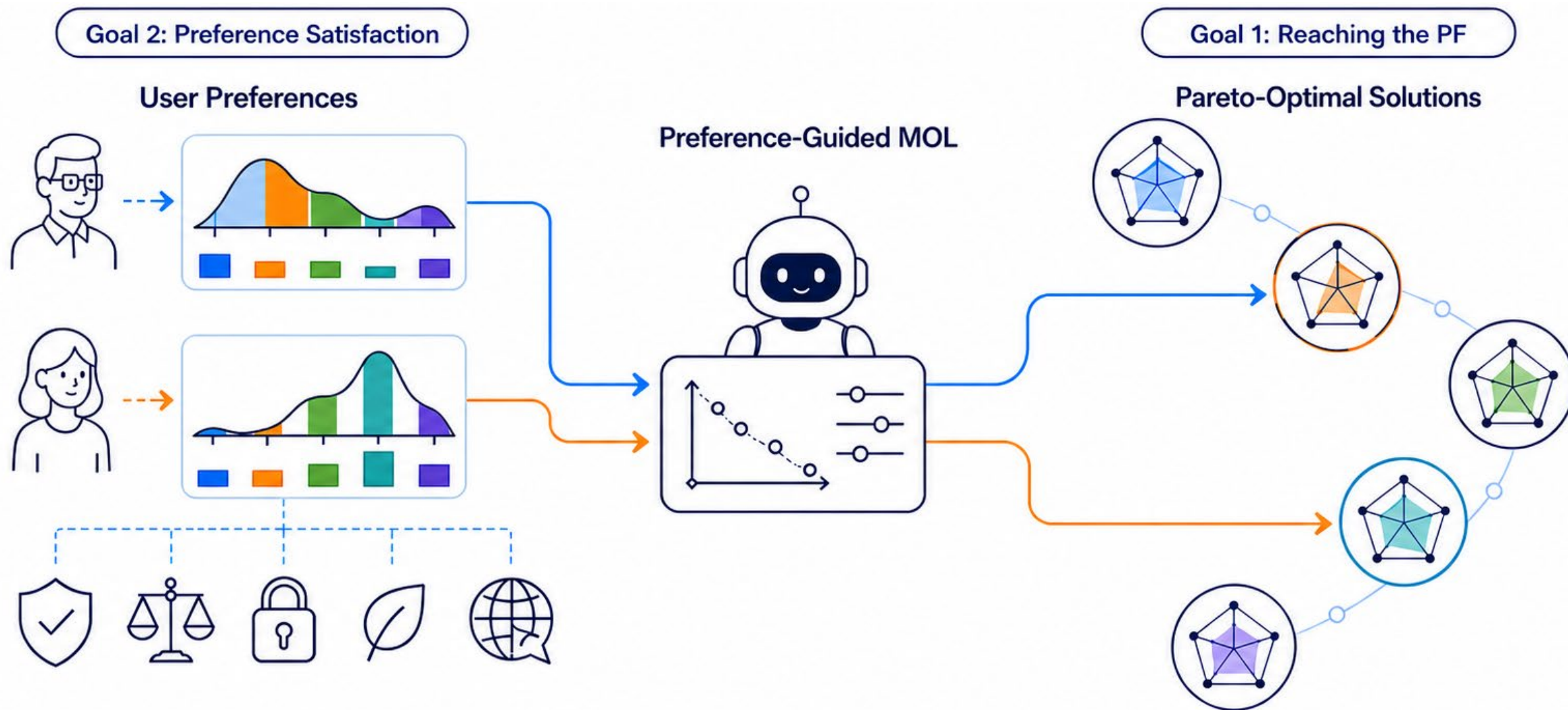
A point $x^* \in X$ is weak Pareto optimal iff there exists no other point $x \in X$ that $F(x) < F(x^*)$.

Definition (Pareto front)

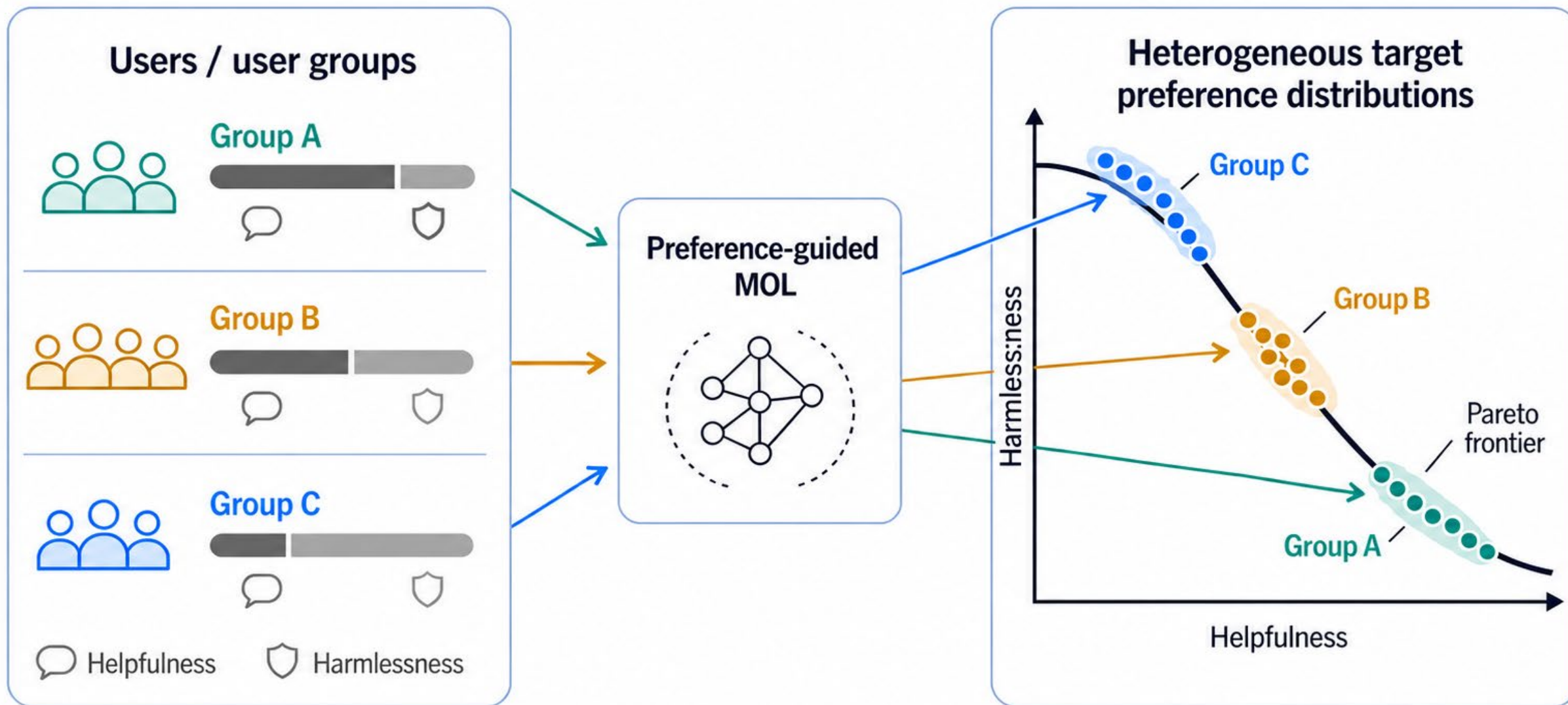
A point y lies on the Pareto front iff $y = F(x^*)$ for some Pareto optimal point x^* . The Pareto front is a collection of $F(x^*)$ for all Pareto optimal point x^* .



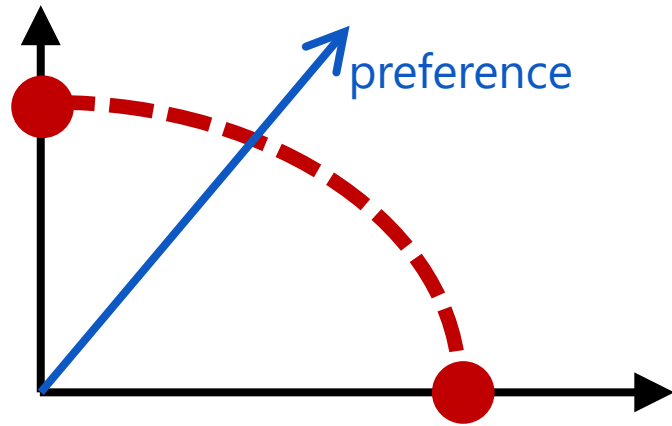
Preference-guided MOL



Heterogeneous target preference distributions



Preference modeled by scalarization weights



By weights: e.g., linear scalarization (LS), Tchebycheff scalarization (TS).

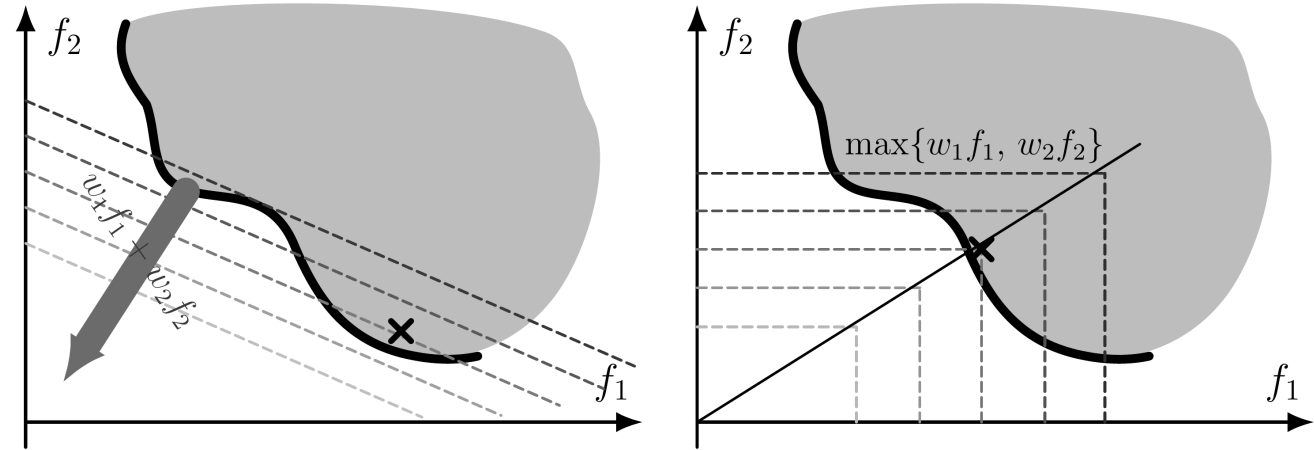
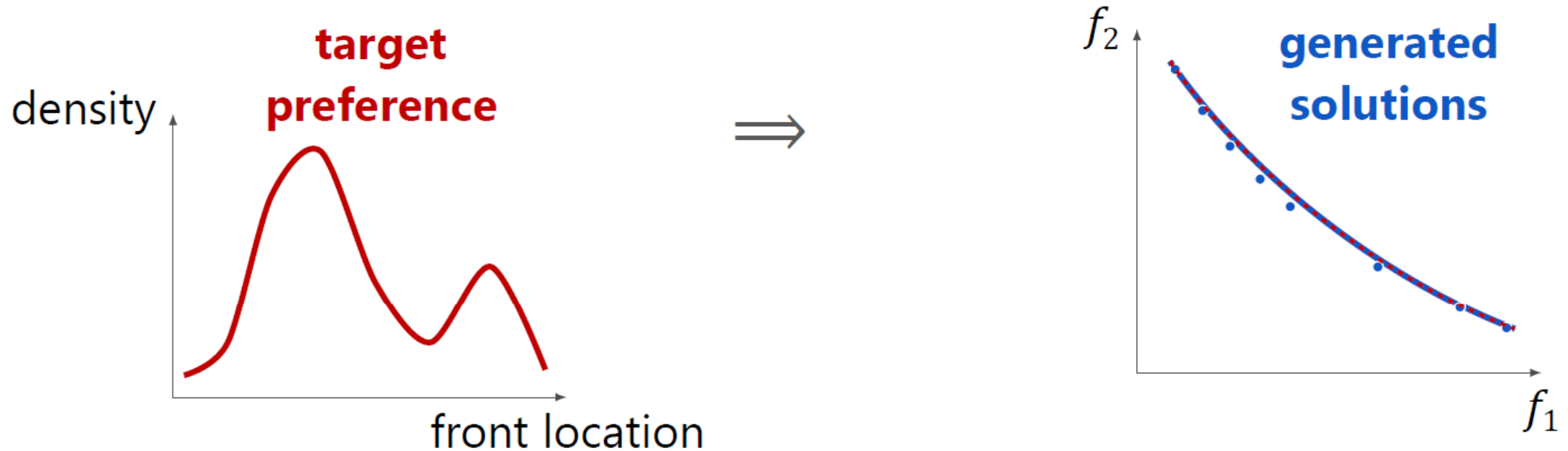


Figure from [Súkeník, 2022]

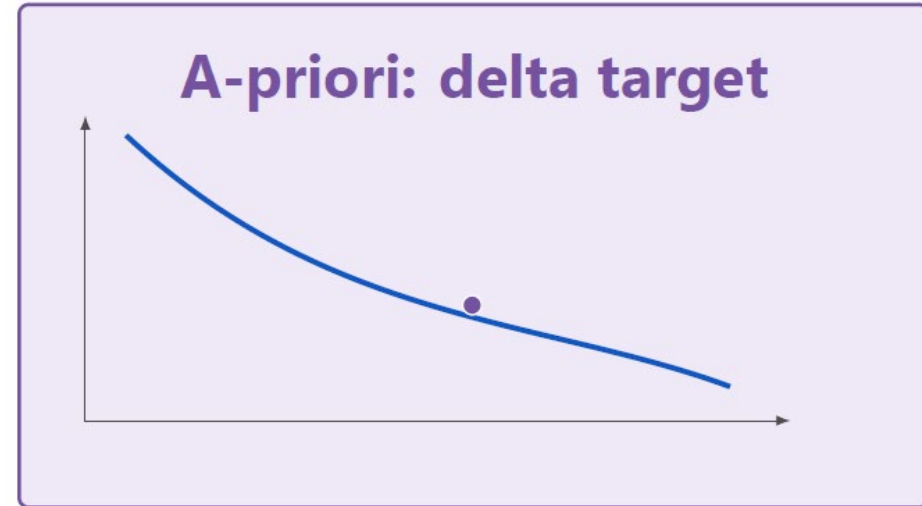
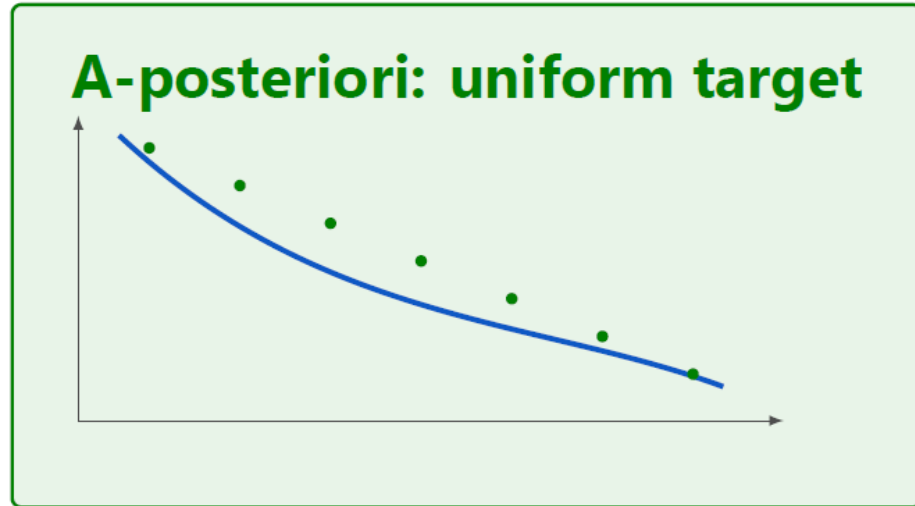
Different **weight** could lead to different solutions on the PF, thus represent different **preferences/trade-offs**

Our goal: distribution matching on PF



What is the desired distribution of scalarization weights to generate the target distribution on the PF?

Two extreme cases: uniform and delta target



Focus of this work

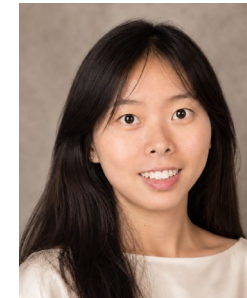
Uniform and delta are two extreme distributions.
We provide the unified distribution matching viewpoint.

Our focus – Uniform coverage (a-posteriori):

First build a balanced uniform menu of trade-offs on the PF,
then let the user decide.



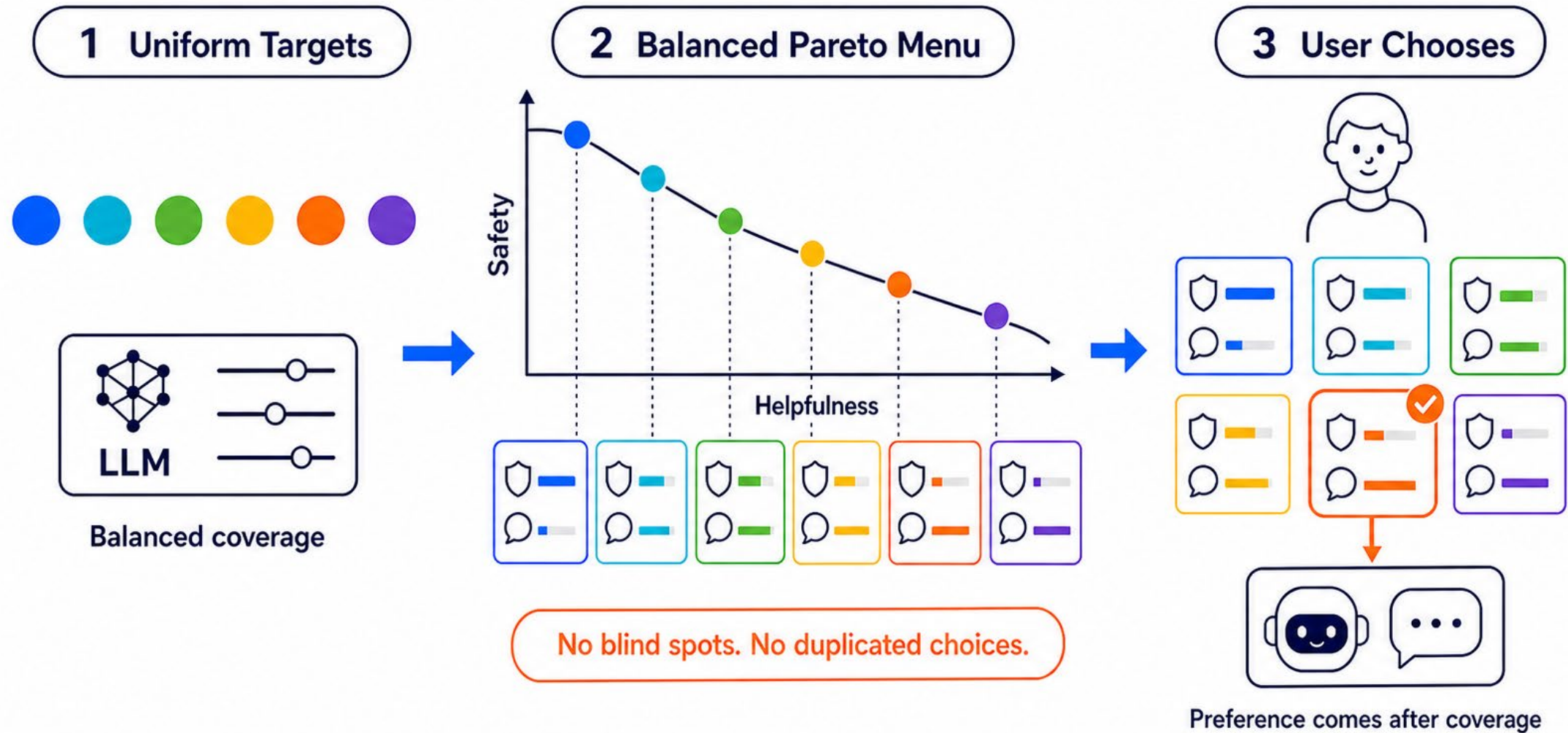
Liuyuan Jiang
U of R



Chentong Huang
U of R

Jiang et al., "SURF: Steering the Scalarization Weight to Uniformly Traverse the Pareto Front," *NeurIPS 2026*.

Why uniform target distribution?



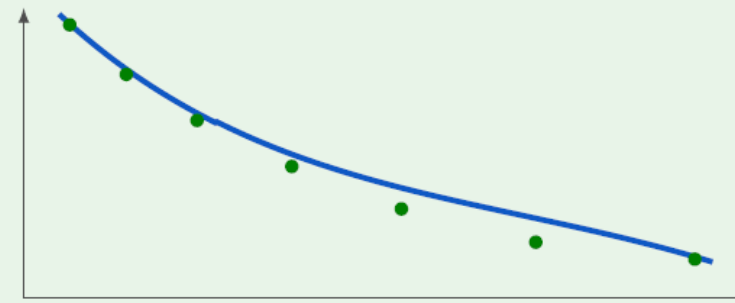
What happens if we sample weights uniformly?

A misleading menu



Seven models, but
few distinct choices

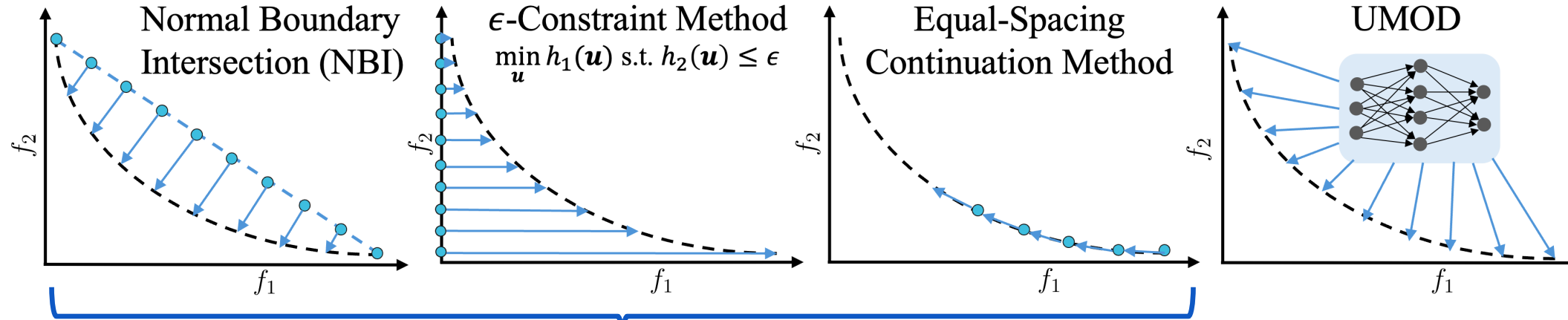
A useful menu



Each model reveals
a new trade-off

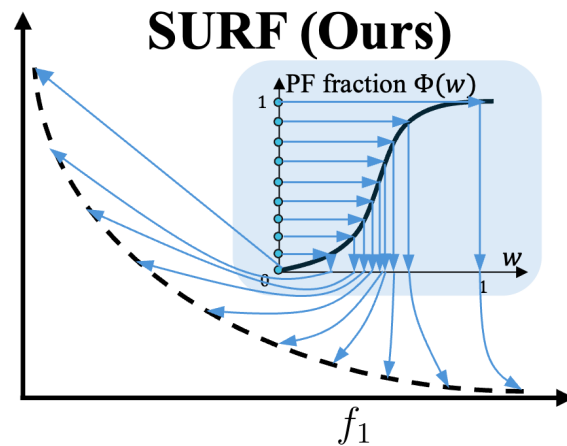
How should we choose weights?

Prior art to address the uniform coverage



Solve constrained opt problem

Find cover to maximize the spread/diversity



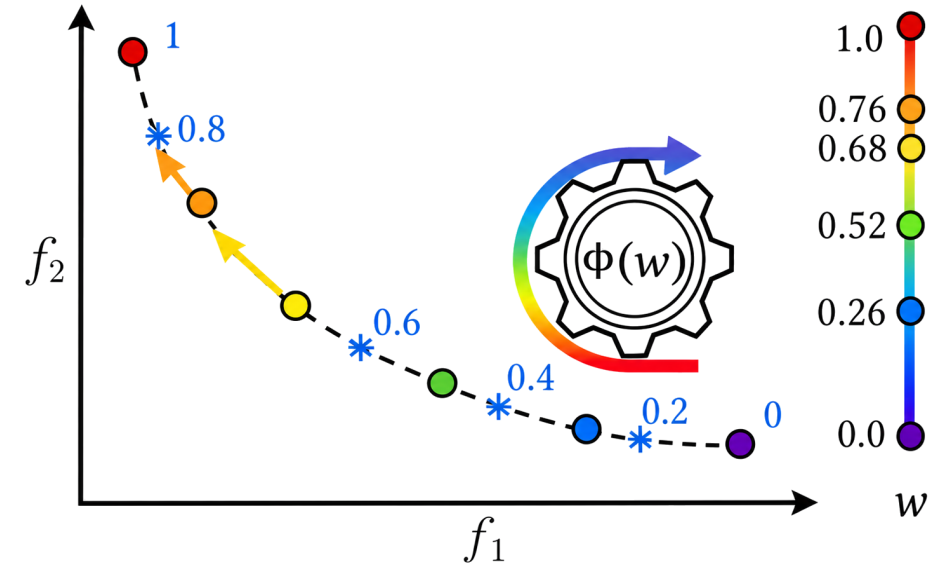
keep the same scalarization solver and choose better weights

Intuitive view: scalarization as a dial

For two objectives, linear scalarization solves

$$x_w^* \in \arg \min_x \{w f_1(x) + (1 - w) f_2(x)\}.$$

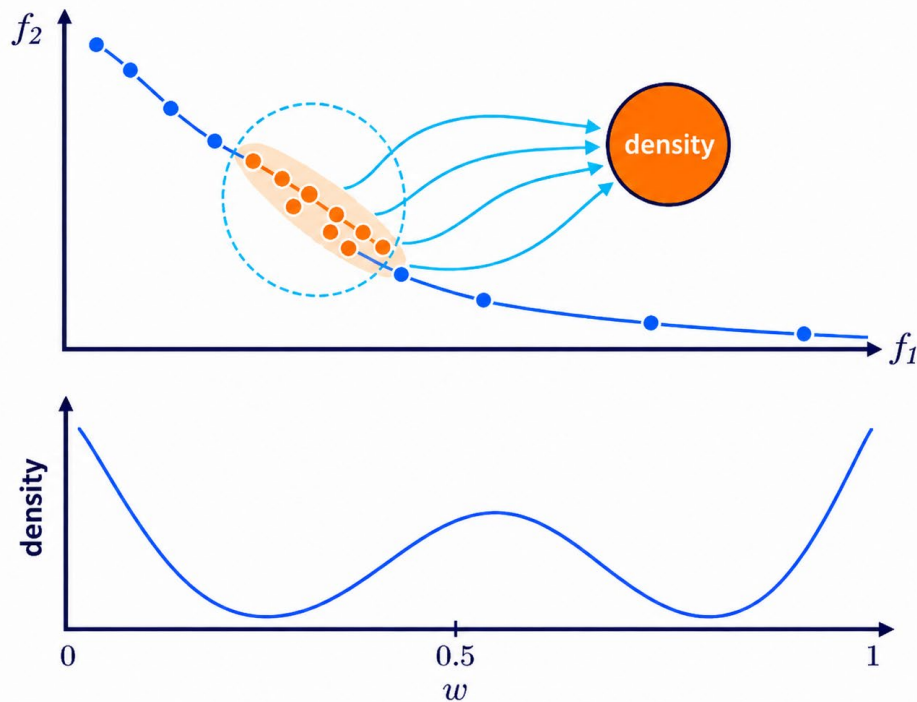
- ▶ $w = 0$: care about objective 2.
- ▶ $w = 1$: care about objective 1.
- ▶ Sweep w to obtain a portfolio of models.



Two views: density and speed

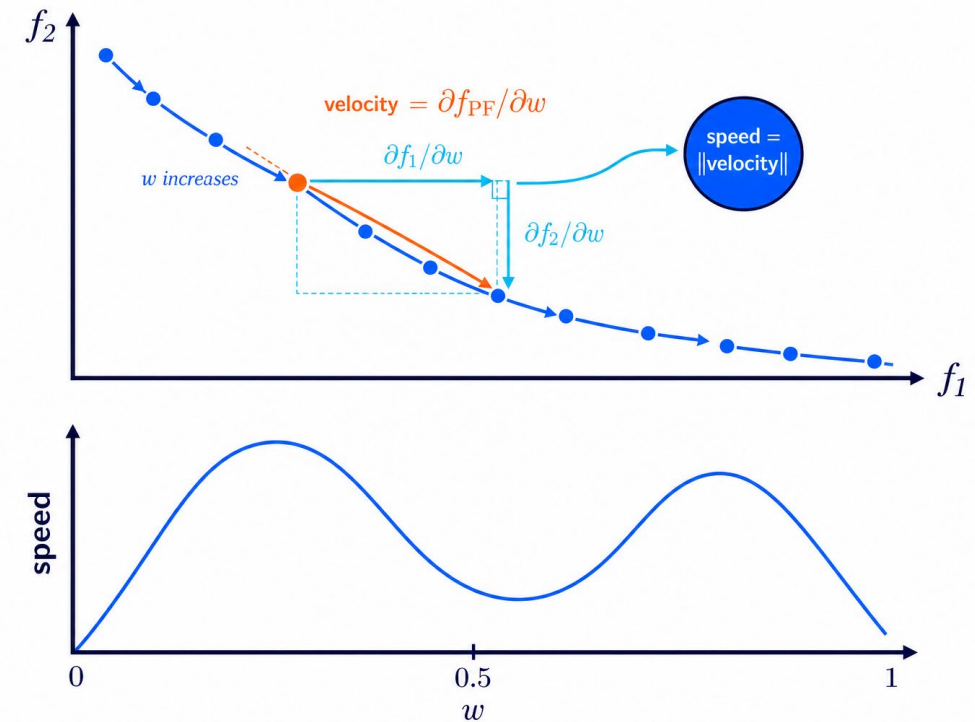
Density view

A scalar: how crowded



Velocity/speed view

A vector: magnitude+direction



Now we take the speed (velocity) views to further illustrate

The geometry: the dial moves at uneven speed

Scalarization-induced path:

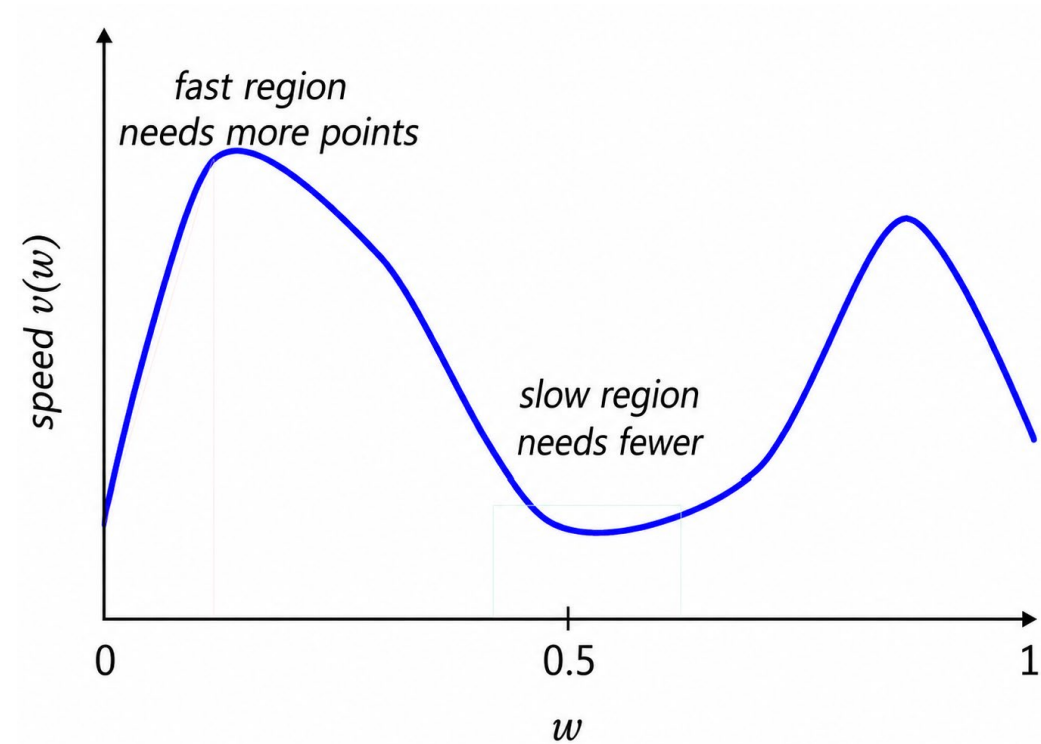
$$F_{\text{PF}}(w) := h(u_w^*) = F(x_w^*).$$

Speed on the front:

$$v(w) = \left\| \frac{\partial}{\partial w} F_{\text{PF}}(w) \right\|.$$

Core mismatch

Uniform weights sample the **dial**; SURF wants to sample **distance along the front**.



SURF turns the dial using Φ^{-1}

Normalize arc length:

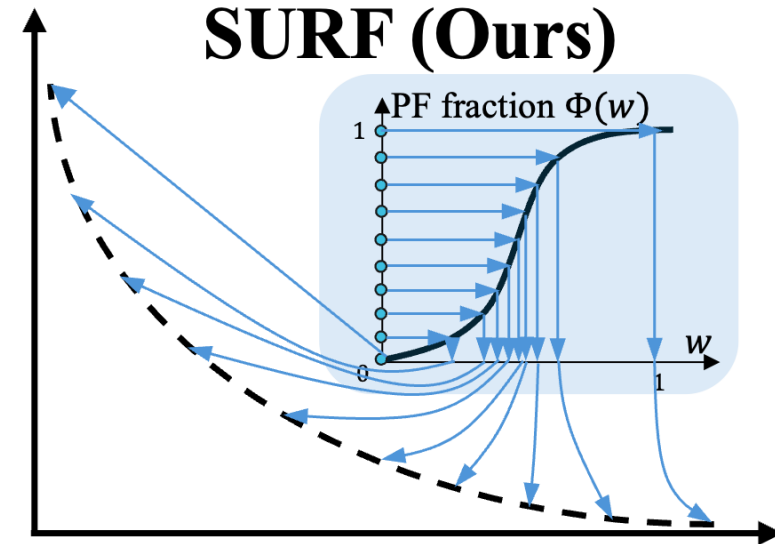
$$\Phi(w) = \frac{s(w)}{s(1)}.$$

PF-aware sampling rule

If Φ is known,

$$w_n = \Phi^{-1}\left(\frac{n}{N}\right).$$

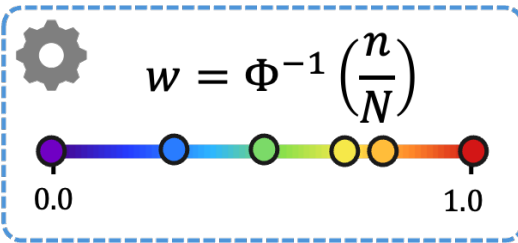
Choose weights so equal steps in n/N become equal **walking distance** on the Pareto front.



Two SURF rules

Rule 1: If Φ available

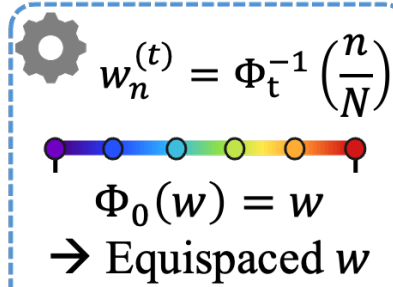
$w \sim \Phi^{-1}$ gives uniform PF



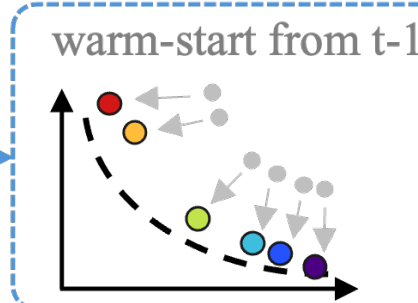
Example (Section 2.1)
close-form $\hat{\Phi}$ in Bandit

Rule 2: General case (Algorithm 1): update $\Phi_t \rightarrow \Phi$

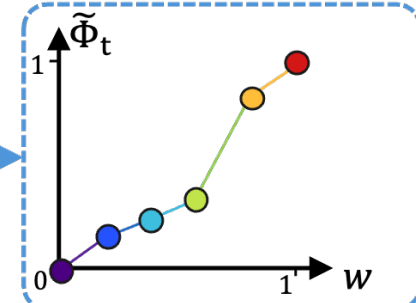
1. Sampling



2. InnerSolver



3. Estimate Φ via $\tilde{\Phi}_t$



$$\text{Update } \Phi_{t+1} = \alpha \tilde{\Phi}_t + (1 - \alpha) \Phi_t$$

Linear Convergence

Rule 1: known Φ

Use inverse-CDF sampling: $w_n = \Phi^{-1}(n/N)$.

Rule 2: unknown Φ

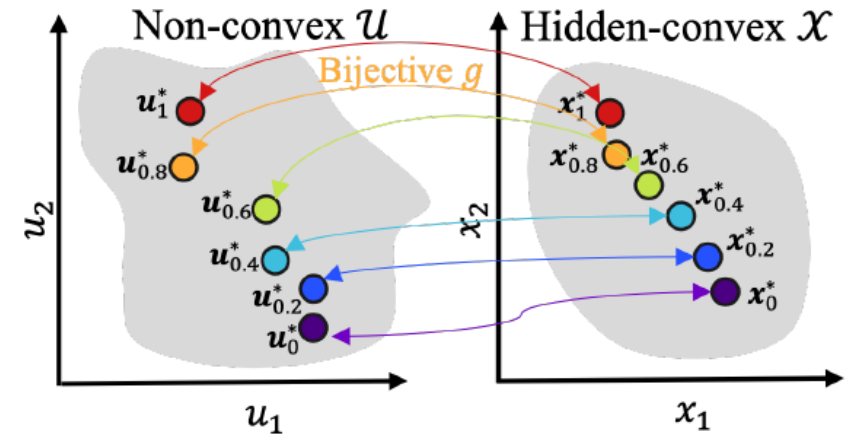
Iterate: solve LS, reconstruct $\hat{\Phi}_t$, then relax toward $\hat{\Phi}_t^{-1}(n/N)$.

When is the scalarization path well-defined?

The analysis is carried out in a hidden convex representation:

$$h_m(u) = f_m(g(u)), \quad x = g(u).$$

$$x_w^* \in \arg \min_{x \in \mathcal{X}} \sum_{m=1}^M w_m f_m(x).$$



- g maps original parameters to hidden space.
- Hidden strong convexity gives a unique smooth path.
- The path maps back to the Pareto front.

Rule 1 with closed form: bandits

For the **bi-objective bandit** case (single state):

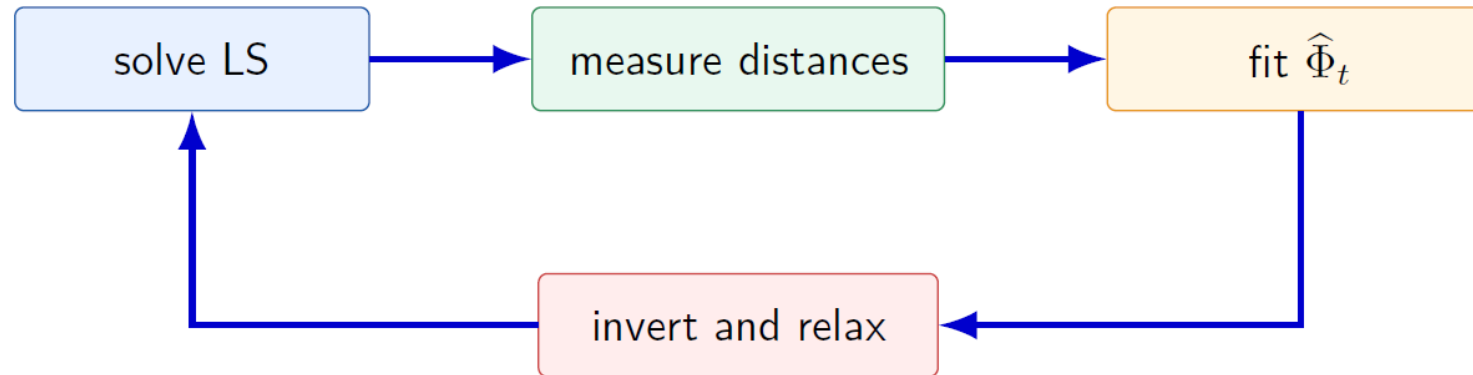
CDF map Φ has a closed form,
thus SURF can be instantiated analytically.

$$v(w) = \left\| \nabla f(x_w^*) \frac{dx_w^*}{dw} \right\|, \quad \Phi(w) = \frac{\int_0^w v(p) dp}{\int_0^1 v(p) dp}.$$

Closed form Φ means no outer learning loop: just invert and solve.

Rule 2 for general case: solve LS and learn map Φ jointly

In most problems, Φ is unknown,
so SURF reconstructs it from sampled points.



What stays the same

Any scalarized optimizer can be used as the inner solver.

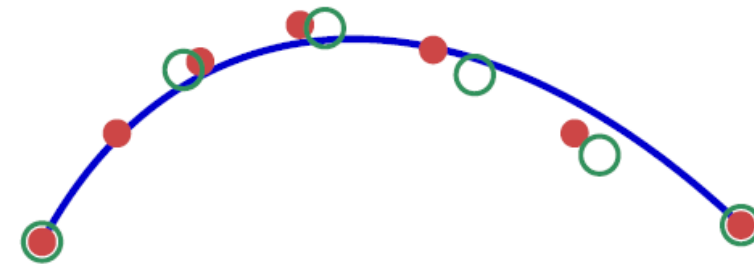
What changes

The outer loop keeps improving the weight schedule.

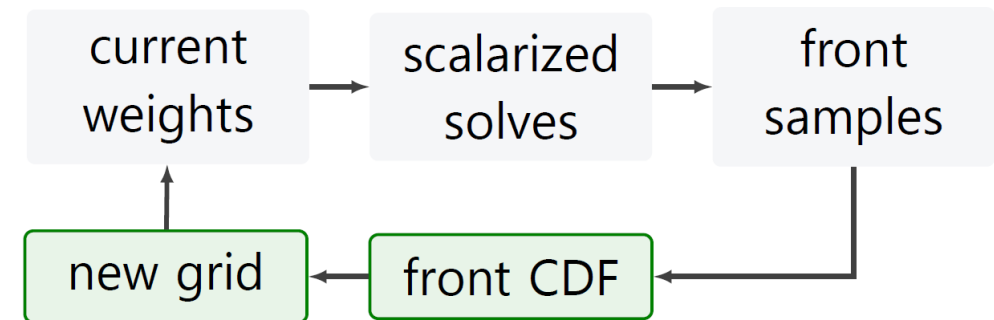
The SURF algorithm under Rule 2

Algorithm 1 SURF, compressed

- 1: initialize $w_n^{(0)} = n/N$
 - 2: **for** $t = 0, \dots, T - 1$ **do**
 - 3: solve scalarized problems at $w_n^{(t)}$
 - 4: reconstruct $\hat{\Phi}_t$ from front distances
 - 5: set $\tilde{w}_n = \hat{\Phi}_t^{-1}(n/N)$
 - 6: relax: $w_n^{(t+1)} = (1 - \alpha)w_n^{(t)} + \alpha\tilde{w}_n$
 - 7: **end for**
-



red: current grid
green: refined grid

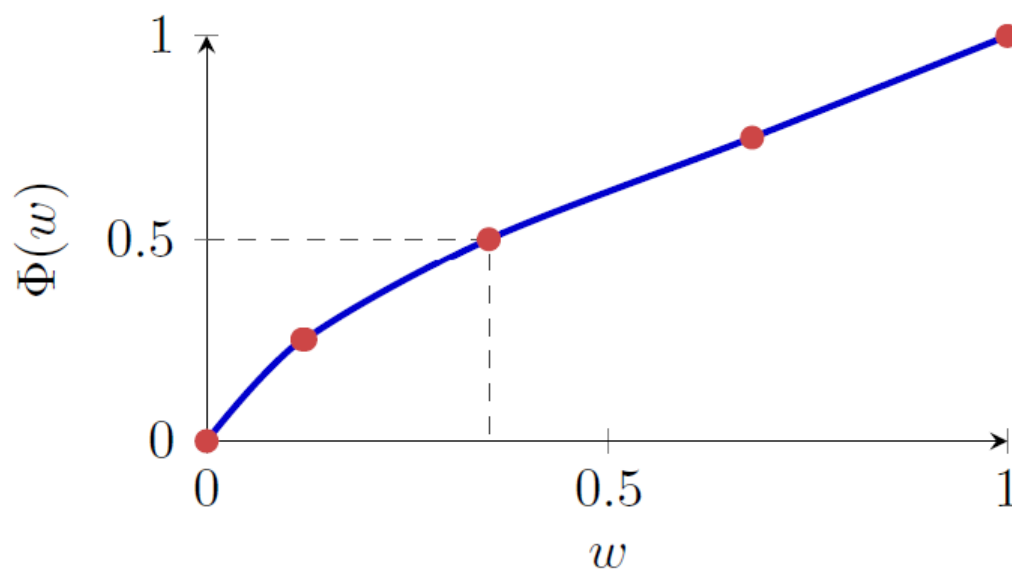


SURF is a scalarization wrapper: it teaches scalarization
where to look next (what weights to sample).

CDF reconstruction

Given objective points ordered by weight, compute piecewise arc lengths:

$$\hat{s}_n = \sum_{i=0}^{n-1} \|h(u_{i+1}) - h(u_i)\|, \quad \hat{\Phi}(w_n) = \frac{\hat{s}_n}{\hat{s}_N}.$$



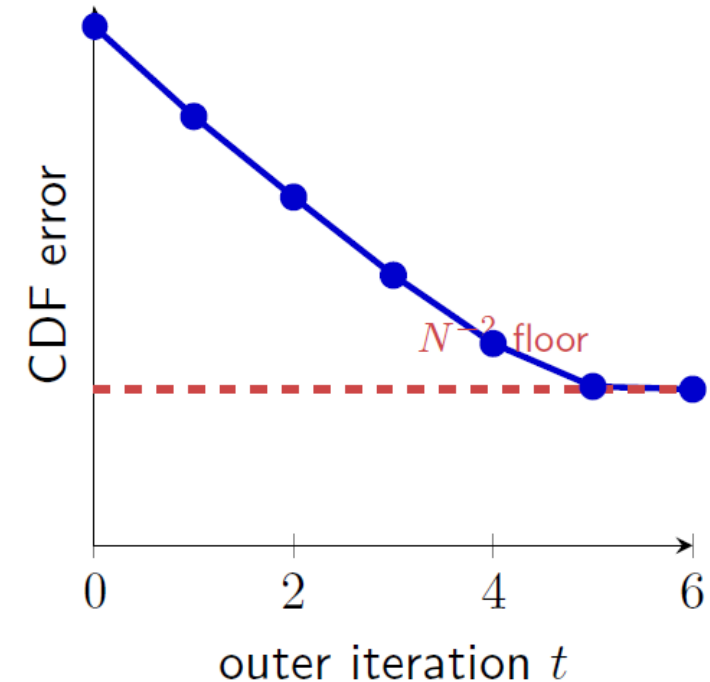
Main theorem: convergence to the sampling floor

Informal guarantee

Under smoothness, nonzero speed, accurate inner solves, and a suitable stepsize,

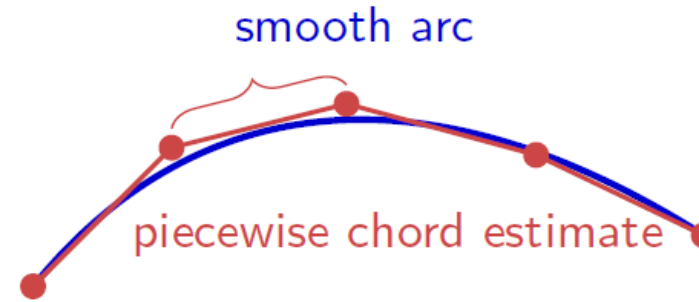
$$\|\Phi_t - \Phi\|_\infty \leq \rho^t \|\Phi_0 - \Phi\|_\infty + CN^{-2}, \quad 0 < \rho < 1.$$

- ▶ ρ^t : outer refinement reduces error.
- ▶ CN^{-2} : finite grid reconstruction limit.



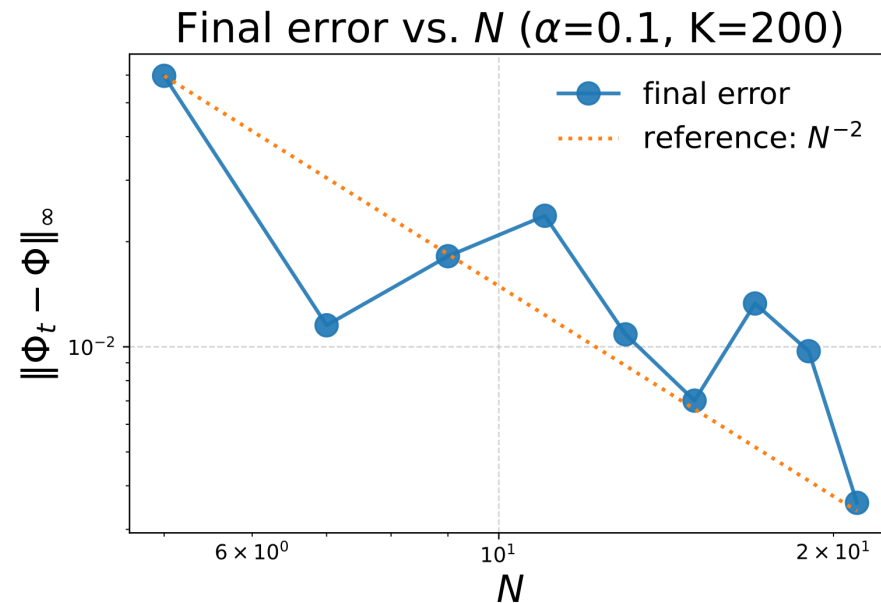
Why the N^{-2} floor is unavoidable

- ▶ The CDF is reconstructed from finite points on the PF.
- ▶ Chords approximate arc length to second order under smoothness.
- ▶ The algorithm can converge only to this discretization floor.



Interpretation

More front points improve the geometric ruler; the theorem says SURF reaches that ruler's accuracy.



Experiment results of SURF

Settings: MO-Gymnasium RL, and LLM alignment

- ▶ Baselines: uniform weights, OLS, UMOD, and reward soup.
- ▶ Metrics: HV, IGD, CV, and gap ratio.

HV: hypervolume

IGD: inverted generational distance

CV: coefficient of variation

MO-Gym: Deep Sea Treasure (DST)

Grid-world setup

A submarine starts at the upper-left and moves with four actions: up, down, left, right.

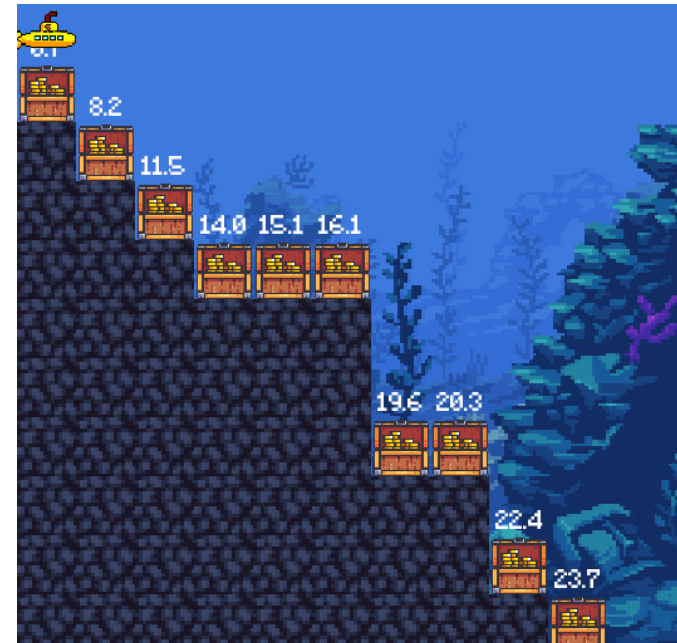
- ▶ State: current grid position (x, y) .
- ▶ Termination: reaching a treasure.
- ▶ Reward vector:
 $r = (\text{treasure value}, -\text{time cost})$.

Trade-off

Fast/shallow treasures give low value; slow/deep treasures give high value.

Why SURF cares

The PF is a set of treasure/time trade-offs, so uniform weights can miss or clump along the front.

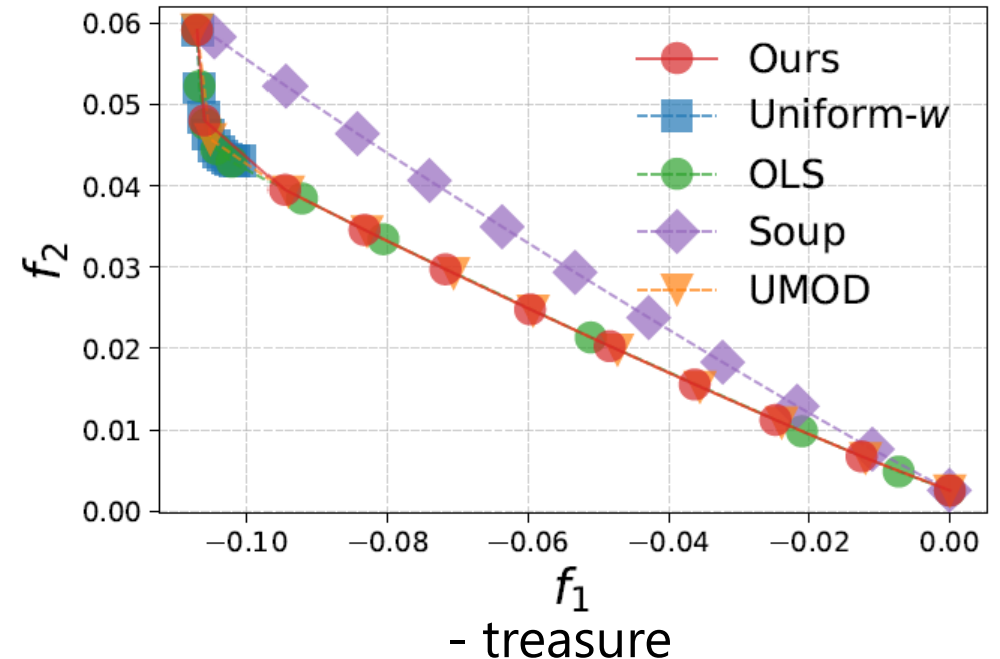


Results on DST: SURF obtains better uniformity

- ▶ Uniform weights often clump solutions on curved regions of the PF.
- ▶ SURF produces more even spacing while preserving or improving PF quality.
- ▶ Gains are visible under fixed total optimization budgets.

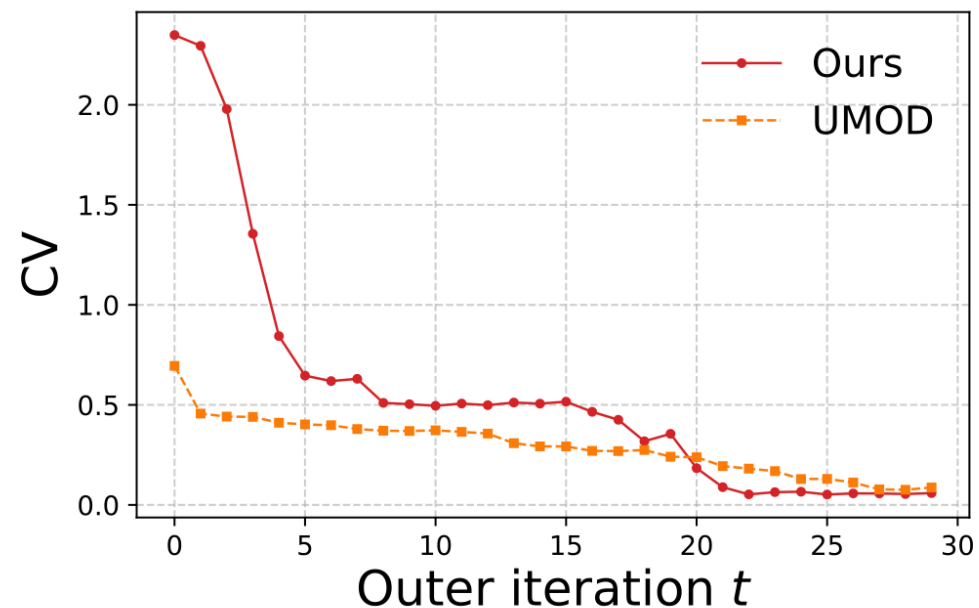
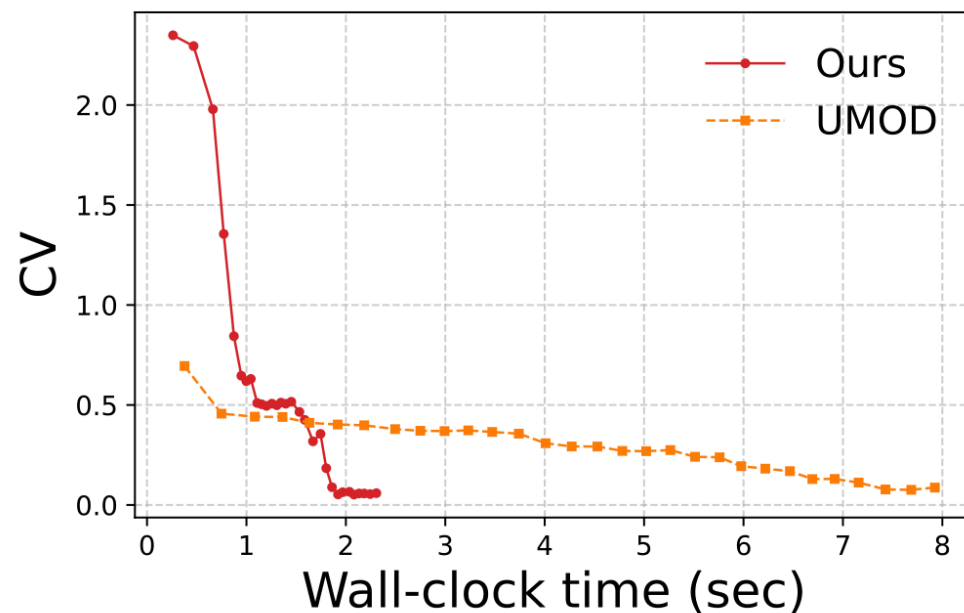
Example reported on Deep Sea Treasure (DST):
SURF improves hypervolume and sharply reduces IGD, CV, and gap ratio versus uniform weighting.

time cost



Results on DST: SURF is more efficient

Convergence comparison between SURF & UMOD on DST.



Results on LLM alignment: SURF has better uniformity

Reddit summarization results (2 objs: helpfulness & harmlessness)

Metric	Soup	Uniform- w	Ours
HV $\uparrow$	0.75 ± 0.30 (0.95 ± 0.30)	3.76 ± 0.31 (4.00 ± 0.30)	4.03 ± 0.35 (4.24 ± 0.35)
CV $\downarrow$	1.69 ± 0.26 (1.69 ± 0.26)	0.65 ± 0.12 (0.65 ± 0.12)	0.30 ± 0.17 (0.29 ± 0.17)
Gap Ratio $\downarrow$	161.20 ± 15.27 (159.83 ± 15.16)	5.89 ± 1.15 (6.01 ± 1.21)	2.93 ± 5.35 (2.89 ± 5.19)

good uniformity

KL-regularized objectives; mean $\pm$ standard deviation over 10 batches.

Takeaway on SURF

Before

uniform in weight space

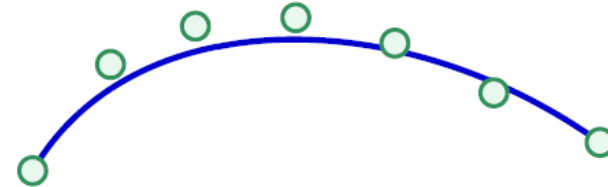
$$w_n = n/N$$



After

uniform in front arc length

$$w_n = \Phi^{-1}(n/N)$$



Estimate the arc-length CDF, invert it, and let scalarization walk evenly.

Treat scalarization as a **pushforward distribution from weights to PF arc length**, then match that distribution to the target one.

References

- L. Jiang, C. Huang, and L. Chen. “SURF: Steering the Scalarization Weight to Uniformly Traverse the Pareto Front,” *Proc. of NeurIPS*, 2026.
- L. Chen, Q. Xiao, E. H. Fukuda, X. Chen, K. Yuan, and T. Chen. “Efficient preference-guided multi-objective learning through first-order optimization on the Pareto set,” *Proc. of ICML*, 2025.
- W. Chen, X. Zhang, B. Lin, X. Lin, H. Zhao, Q. Zhang, and J. T. Kwok. “Gradient-based multi-objective deep learning: Algorithms, theories, applications, and beyond,” *arXiv preprint arXiv:2501.10945*, 2025.
- L. Chen, A. F. M. Saif, Y. Shen, and T. Chen. “FERERO: A flexible framework for preference-guided multi-objective learning,” *Proc. of NeurIPS*, 2024.
- L. Chen*, H. Fernando*, Y. Ying, T. Chen. “Three-way trade-off in multi-objective learning: Optimization, generalization and conflict-avoidance,” *JMLR*, 2024, short version at *NeurIPS*, 2023.
- X. Zhang, G. Li, X. Lin, Y. Zhang, Y. Chen, and Q. Zhang. “Gliding over the Pareto front with uniform designs,” *Proc. of NeurIPS*, 2024.
- A. Ramé, G. Couairon, M. Shukor, C. Dancette, J.-B. Gaya, L. Soulier, and M. Cord. “Rewarded soups: Towards Pareto-optimal alignment by interpolating weights fine-tuned on diverse rewards,” *Proc. of NeurIPS*, 2023.

References

- L. Barrett and S. Narayanan. "Learning all optimal policies with multiple criteria," *Proc. of ICML*, 2008.
- I. Das and J. E. Dennis. "Normal-boundary intersection: A new method for generating the Pareto surface in nonlinear multicriteria optimization problems," *SIAM Journal on Optimization*, 1998.
- S. Liu and L. Nunes Vicente. "The stochastic multi-gradient algorithm for multi-objective optimization and its application to supervised machine learning," *Annals of Operations Research*, 2021.
- D. Mahapatra and V. Rajan. Multi-task learning with user preferences: Gradient descent with controlled ascent in Pareto optimization. *Proc. of ICML*, 2020.
- X. Lin, H.-L. Zhen, Z. Li, Q.-F. Zhang, and S. Kwong. "Pareto multi-task learning," *Proc. of NeurIPS*, 2019.
- O. Sener and V. Koltun. "Multi-task learning as multi-objective optimization," *Proc. of NeurIPS*, 2018.
- P. Skenk and C. H. Lampert. "Generalization in Multi-Objective Machine Learning," *arXiv preprint arXiv:2208.13499*, 2022.
- E. H. Fukuda and L. M. G. Drummond. "A survey on multi-objective descent methods," *Pesquisa Operacional*, 2014.
- J. Fliege and B. F. Svaiter. "Steepest descent methods for multicriteria optimization," *Mathematical methods of operations research*, 2000.